

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 22/03/2026**
Time: 3 Hrs **GTM - 01** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A force is represented by $F = ax^2 + bt^{\frac{1}{2}}$ where $x = \text{distance}$ and $t = \text{time}$. The dimensions of $\frac{b^2}{a}$:

A) $\left[ML^2T^{-3}\right]$ B) $\left[ML^3T^{-3}\right]$ C) $\left[ML^{-2}\right]$ D) $\left[ML^{-1}T^{-1}\right]$

2. A parallel-plate capacitor (plate area A, separation d) is connected to a battery of emf V. The space between the plates is filled with a dielectric whose dielectric constant varies as $K(x) = 1 + \frac{x}{d}$, The electric filed is uniform: $E = \frac{V}{d}$. What is the average energy density stored in the capacitor?

A) $\frac{\varepsilon_0 V^2}{2d^2}$ B) $\frac{3\varepsilon_0 V^2}{4d^2}$ C) $\frac{\varepsilon_0 V^2}{d^2}$ D) $\frac{2\varepsilon_0 V^2}{d^2}$

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3. What is the relative decrease in the focal length of a lens when its optical power is increased by 0.2 D from 4.0 D?

A) 0.08 B) 0.05 C) 0.20 D) 0.10

4. If the ratio of number of moles of hydrogen gas to oxygen gas is two, the ratio of total translational kinetic energy of hydrogen gas molecules to that of oxygen molecules at $20^0 C$ is__

A) 1:1 B) 2:1 C) 1:8 D) 1:16

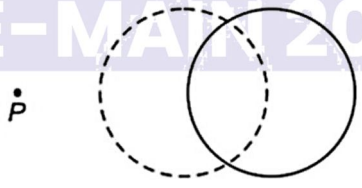
5. A plano-convex lens having radius of curvature of first surface 2 cm exhibits focal length of f_1 in air. Another plano-convex lens with first surface radius of curvature 3 cm has focal length of f_2 when it is immersed in a liquid of refractive index 1.2. If both the lenses are made of same glass of refractive index 1.5, the ratio of f_1 and f_2 will be:-

A) 3:5 B) 1:3 C) 1:2 D) 2:3

6. Two identical coaxial rings each of radius R are separated by a distance of $\sqrt{3}$ R. They are uniformly charged with charges +Q and –Q respectively. The minimum kinetic energy with which a charged particle (charge +q) should be projected from the centre of the negatively charged ring along the axis of the rings such that it reaches the centre of the positively charged ring is

(A) $\frac{Qq}{4\pi\varepsilon_0 R}$ (B) $\frac{3Qq}{2\pi\varepsilon_0 R}$ (C) $\frac{Qq}{2\pi\varepsilon_0 R}$ (D) $\frac{3Qq}{4\pi\varepsilon_0 R}$

7. Figure shows a closed dotted surface which intersects a conducting uncharged sphere. If a positive charge is placed at the point P, the flux of the electric field through the closed surface



(A) will remain zero (B) will become positive

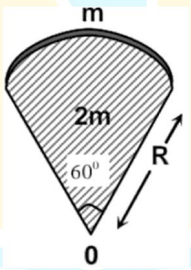
(C) will become negative (D) data insufficient

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8. A gas is found to be obeyed the law p^2V = constant. The initial temperature and volume are T_0 and V_0. If the gas expands to a volume $3V_0$, then the final temperature becomes

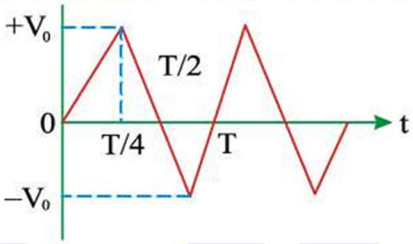
(A) $\sqrt{3}T_0$ (B) $\sqrt{2}T_0$ (C) $\frac{T_0}{\sqrt{3}}$ (D) $\frac{T_0}{\sqrt{2}}$

9. What is the distance of centre of mass of the combined system as shown in figure from 'O'. System consists of a uniform sector of mass 2m and radius R, fixed with a uniform arc of mass m and radius R along the edge of the sector. (System is making an angle θ = 60° at their common centre)



(A) $\frac{5R}{3\pi}$ (B) $\frac{4R}{3\pi}$ (C) $\frac{7R}{3\pi}$ (D) none of these

10. The voltage time $(V-t)$ graph for triangular wave having peak value V_0 is shown in figure.

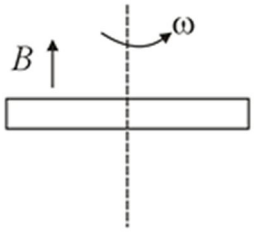


The rms value of V in time interval from t = 0 to $\frac{T}{4}$ is

A) $\frac{V_0}{\sqrt{3}}$ B) $\frac{V_0}{2}$ C) $\frac{V}{\sqrt{2}}$ D) $2V_0$

11. A conducting rod of length $2l$ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The emf induced between two ends of the rod is

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(A) $B\omega l^2$ (B) $\frac{1}{2}B\omega l^2$ (C) $\frac{1}{8}B\omega l^2$ (D) Zero

12. The energy that should be added to an electron, to reduce its de-Broglie wavelengths from 10^{-10}m to 0.5×10^{-10} m, will be

(A) four times the initial energy (B) thrice the initial energy

(C) equal to the initial energy (D) twice the initial energy

13. The wavelength of a certain line in the x-ray spectrum for tungsten (Z= 74) is 200 Å. What would be the wavelength of the same line for platinum (Z= 78)? The screening constant a is unity.

(A) 179.76 Å (B) 189.76 Å (C) 289.76 Å (D) 379.76 Å

14. A ray of light travelling in a medium of refractive index μ is incident at an angle θ on a composite transparent plate consisting of 50 plates of refractive indices 1.01 μ, 1.02 μ, 1.03 μ, …., 1.50 μ. The ray emerges from the composite plate into a medium of refractive index 1.6 μ at angle x. Then

(A) $\sin x = \left(\frac{1.01}{1.5}\right)^{50} \sin\theta$ (B) $\sin x = \frac{5}{8}\sin\theta$

(C) $\sin x = \frac{8}{5}\sin\theta$ (D) $\sin x = \left(\frac{1.5}{1.01}\right)^{50} \sin\theta$

15. The binding energy per nucleon of deuteron $\left({}_1^2\mathrm{H}\right)$ and helium nucleus $\left({}_2^4\mathrm{He}\right)$ is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is

(A) 23.6 MeV (B) 26.9 MeV (C) 13.9 MeV (D) 19.2 MeV

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16. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. Almost instantaneously, another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector ?

(A) two photons of energy 10.2 eV

(B) two photons of energy 1.4 eV

(C) one photon of energy 10.2 eV and an electron of energy 1.4 eV

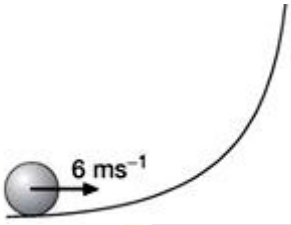
(D) one electron having kinetic energy nearly 11.6 eV

17. Two electrons are moving with non-relativistic speeds perpendicular to each other. If corresponding de Broglie wavelengths are λ_1 and λ_2, their de Broglie wavelength in the frame of reference attached to their centre of mass is

A) $\lambda_{CM} = \dfrac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$ B) $\lambda_{CM} = \lambda_1 = \lambda_2$

C) $\lambda_{CM} = \left(\dfrac{\lambda_1 + \lambda_2}{2}\right)$ D) $\dfrac{1}{\lambda_{CM}} = \dfrac{1}{\lambda_1} + \dfrac{1}{\lambda_2}$

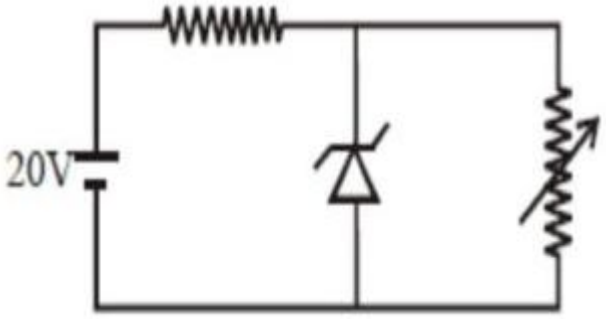
18. A disc of radius 0.1 m rolls without sliding on a horizontal surface with a velocity of $6\,ms^{-1}$. It then ascends a smooth continuous track as shown in figure. Taking $g = 10\ ms^{-2}$, the height up to which it will ascend is



A) 0.9 m B) 1.8 m C) 2.4 m D) 2.7 m

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19. In the given circuit, the voltage across the load is maintained at 12V. The current in the zener diode varies from 0–50mA. What is the maximum wattage of the diode?



A) 12 w B) 6 w C) 0.6 w D) 1.2 w

20. If a million tiny droplets of water of the same radius coalesce into one larger drop, then the ratio of the surface energy of the large drop to the total surface energy of all droplets will be

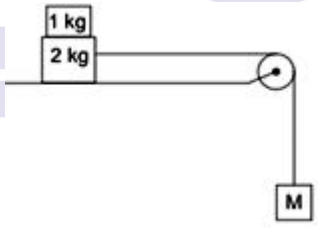
A) 1:10 B) $1:10^2$ C) $1:10^4$ D) $1:10^6$

SECTION – II

(Numerical Value Answer Type)

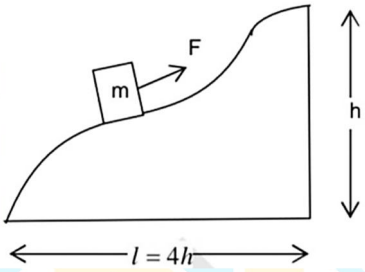
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. Find minimum value of mass suspended M for which 1 kg block start toppling. Assume 1 kg block does not slip over smooth 2 kg block. Length of 1 kg block is 4 cm and height is 8 cm. (ground is smooth).



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22. A body of mass 'm' is slowly hauled up a hill by an external force which is always acting along the surface as shown. The coefficient of friction between the body and the surface is $\mu = 0.5$. If work done by friction and gravity on the body are ΔW_{fr} and ΔW_g respectively during the process. Find the ratio $\frac{\Delta W_{fr}}{\Delta W_g}$.



23. A particle is executing simple harmonic motion with time period 2s and amplitude 1 cm. If D and d are the total distance and displacement covered by the particle in 12.5 s, then $\frac{D}{d}$ is:-

24. An object is displaced from position vector $\vec{r}_1 = \left(2\hat{i} + 3\hat{j}\right)$ m to position vector $\vec{r}_2 = \left(4\hat{i} + 6\hat{j}\right)$ m under a force $\vec{F} = \left(3x^2\hat{i} + 2y\hat{j}\right)$ N. Calculate the work done by this force (in Joule).

25. The least count of a screw gauge is $\frac{1}{100} mm$ and the pitch of the screw is $1 mm$. The maximum percentage error of the instrument is

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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26. Zinc and hydrochloric acid react according to the reaction.

$$Zn(s) + 2HCl(aq.) \rightarrow ZnCl_2(aq.) + H_2(g)$$

If 0.30 mole of Zn are added to hydrochloric acid containing 0.52 mole HCl , how many moles of H_2 are produced?

(A) 0.26 (B) 1.04 (C) 0.52 (D) 0.13

27. The ratio of $(E_2 - E_1)$ to $(E_4 - E_3)$ for He^+ ion is approximately equal to (where E_n is the energy of nth orbit)

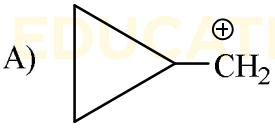
(A) 10 (B) 15 (C) 17 (D) 12

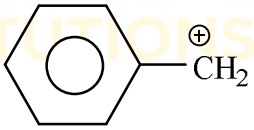
28. Correct order of bond energy is:

(A) $N_2 > N_2^+ > N_2^- > N_2^{2-}$ (B) $N_2^+ > N_2^- > N_2^{2-} > N_2$

(C) $N_2 > N_2^- = N_2^+ > N_2^{2-}$ (D) $N_2^- > N_2 = N_2^+ > N_2^{2-}$

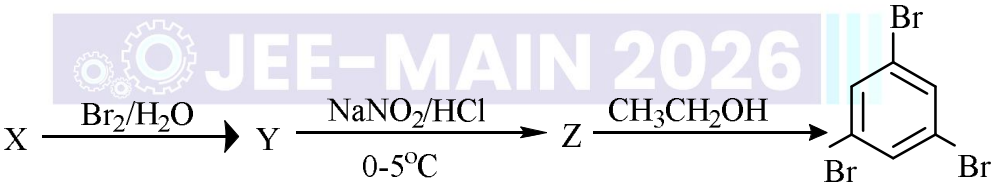
29. Select the least stable carbocation among the following

A) 

B) 

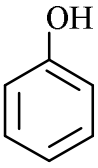
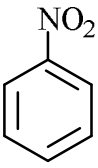
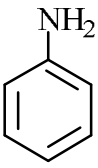
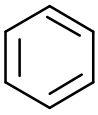
C) $CH_2 = CH - \overset{\oplus}{C}H - NO_2$ D) $CH_2 = CH - \overset{\oplus}{C}H - CH_3$

30. In the reaction sequence.



Identify X

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A)  B)  C)  D) 

31. A solution containing 4 g of polyvinyl chloride in 1 litre of dioxane was found to have an osmotic pressure of 6×10^{-4} atm at 300 K. The molecular mass of polymer is :

(A) 3×10^{3} (B) 1.6×10^{5} (C) 5×10^{4} (D) 6.4×10^{2}

32. In a reversible reaction $A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B$, the initial concentration of A and B are a and b in moles per litre, k_1 and k_2 are rate constants for forward & backward reactions respectively and the equilibrium concentrations are (a – x) and (b+x) respectively; express x in terms of k_1, k_2, a and b.

(A) $\frac{k_1a-k_2b}{k_1+k_2}$ (B) $\frac{k_1a-k_2b}{k_1-k_2}$ (C) $\frac{k_1a-k_2b}{k_1k_2}$ (D) $\frac{k_1a+k_2b}{k_1+k_2}$

33. For the given reaction, identify the true (T) & false (F) statements.

$C_2H_5NH_2 + HI \rightleftharpoons (C_2H_5NH_3)^+ I^-$

S_1 : HI is bronsted base.

S_2 : HI is bronsted acid.

S_3 : HI is arrhenius acid.

S_4 : HI is lewis acid.

S_5 : HI is arrhenius base.

S_6 : HI is lewis base.

(A) T F F F T T (B) F T T T F F

(C) F T T F F F (D) T F F F T F

34. Na-amalgam is prepared by electrolysis of NaCl solution using liquid Hg as cathode. How long should the current of 10 amp. is passed to produce 10% Na – Hg on a cathode of 10 g Hg. (atomic mass of Na = 23).

(A) 7.77 min (B) 9.44 min (C) 5.24 min (D) 11.39 min

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35. The compound MX_4 is tetrahderal. The number of < XMX angles in the compound is

(A) Three (B) Four (C) Five (D) Six

36. The number of moles of NaOH required to completely react with 20 ml of 0.1M NaH_2PO_3 solution and 40 ml of 0.1M $NaHCO_3$.

(A) 8×10^{-3} (B) 2×10^{-3} (C) 4×10^{-3} (D) 6×10^{-3}

37. Group 13 elements exhibit

(A) Only +3 oxidation state (B) Only +1 oxidation state

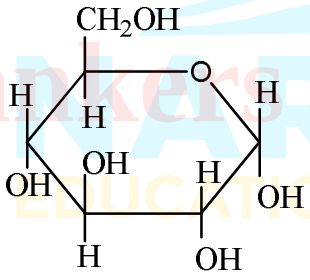
(C) Both +1 and +3 oxidation states (D) +1, +2 and +3 oxidation states.

38. Carbon monoxide is formed by the action of concentrated H_2SO_4 on

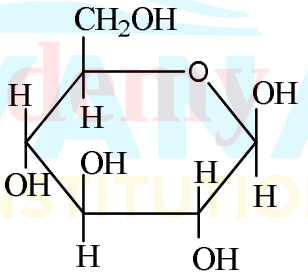
(A) formic acid (B) oxalic acid

(C) potassium ferrocyanide (D) all of these

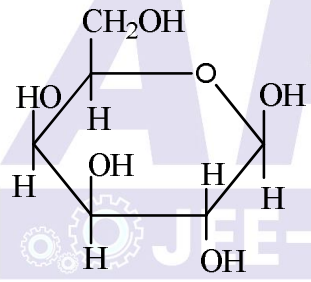
39. Consider the structure of carbohydrates



(I)



(II)



(III)

Select the correct statement

A) I and II are C-4 epimer

B) II and III are anomer

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C) II and III are C-4 epimer

D) I and II are $\alpha-D-$Glucose and $\alpha-D-$Galactose respectively.

40. Consider the following complexes -

(i) K_2PtCl_6 (ii) $PtCl_4\cdot 2NH_3$

(iii) $PtCl_4\cdot 3NH_3$ (iv) $PtCl_4\cdot 5NH_3$

Their electrical conductances in aq. solutions are-

(A) 256, 0, 97, 404 (B) 404, 0, 97, 256

(C) 256, 97, 0, 404 (D) 404, 97, 256, 0

41. Product-I $\xleftarrow{\text{aq. KOH}}$ C_2H_5Br $\xrightarrow{\text{alc KOH}}$ Product-II, the correct statement is

(A) Product-I is obtained by the elimination reaction

(B) Product-II is obtained by the substitution reaction

(C) The molecular formula of Product - I is C_2H_4, while the molecular formula of Product - II is C_2H_6O

(D) Product-I is the isomer of dimethyl ether, while Product-II is the dehydrated compound of Product-I

42. $CH_2=CH-CH=CH_2 \xrightarrow[1\,\text{equiv.}]{H_2/Pt} A \xrightarrow{O_3/H_2O} B$

A and B are

(A) $CH_3-CH_2-CH=CH_2, (CH_3CH_2COOH+CO_2)$

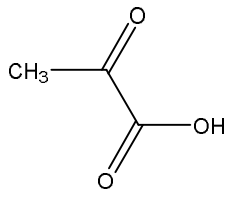
(B) $CH_3-CH=CH-CH_3, CH_3COOH(2\text{ mol})$

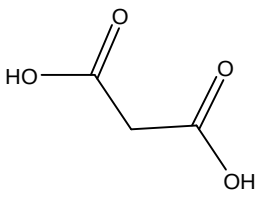
(C) $CH_3-CH=CH-CH_3, CH_3CHO(2\text{ mol})$

(D) $CH_3-CH_2-CH=CH_2, (CH_3CH_2CHO+HCHO)$

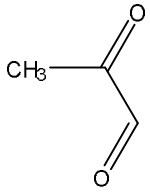
43. End product of the following sequence of reactions is

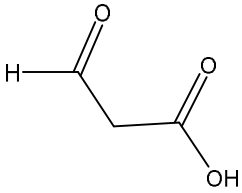
$CH\equiv CH \xrightarrow{CH_3MgBr} \xrightarrow{CO_2/H_3O^+} \xrightarrow{HgSO_4/H_2SO_4} \xrightarrow[\Delta]{Ag_2O}$

(A) 

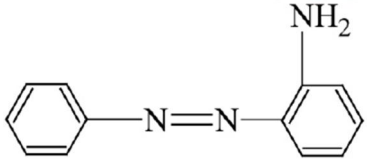
(B) 

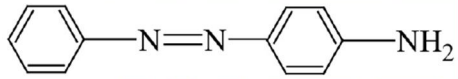
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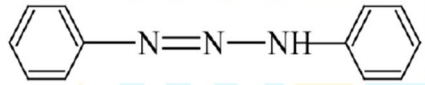
(C) 

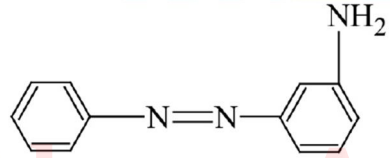
(D) 

44. When aniline is treated with benzene diazonium chloride at low temperature in weakly acidic medium, the final product obtained is

(A) 

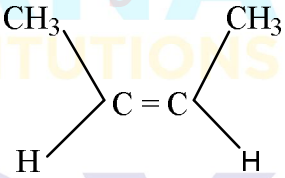
(B) 

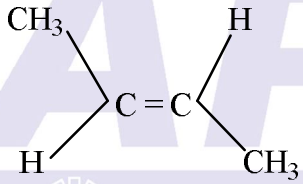
(C) 

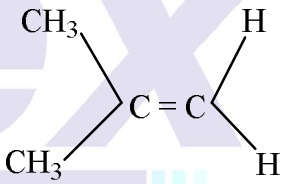
(D) 

45. The compound which reacts with HBr obeying Markownikov's rule is -

(A) $CH_2 = CH_2$

(B) 

(C) 

(D) 

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. 300 ml of 3.0 M NaCl is added to 200 ml of 4.0M $BaCl_2$ solution. The concentration of Cl^- ions in the resulting solution is _____ M.

47. The pK_b of NH_4OH is 4.75. An aqueous solution of NH_4OH is titrated with HCl. The pH of the solution at a point when half of NH_4OH is neutralized is x. Give value equal to 100x.

48. The gaseous potassium chloride molecule has a measured dipole moment of 10.0 D, which indicates that it is a very polar molecule. The separation between the nuclei in this molecule is 2.67×10^{-8} cm. Calculate the percentage ionic character in KCl molecule.

49. A 1 L reaction vessel which is equipped with a movable piston is filled completely with a 1 M aqueous solution of H_2O_2. The H_2O_2 decomposes to $H_2O(l)$ and $O_2(g)$ in a first order process with half life 5 hrs at 300 K. As gas formed, the piston moves up against constant external pressure of 1 atm. What is the net work done (in cal) by the gas from the start of sixth hour till the end of 10 hrs ?

50. 0.257 g of an organic substance was heated with conc H_2SO_4 and then distilled with excess of strong alkali. The ammonia gas evolved was absorbed in 50 ml of N/10 HCl which required 23.2 ml of N/10 NaOH for neutralisation at the end of the process. The percentage of nitrogen in the compound is x % (Rounded off to one decimal place). Give value of 10x.

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If a sequence is given by $\{a_n\}$ where $a_1 = 1$ & $a_{n+1} = \frac{a_n}{1 + n \cdot a_n} (n \in N)$ then value of $\frac{1}{a_{1000}} - 499500$ is

(A) 2 (B) 3 (C) 1 (D) 4

52. If for some values of $a, b, c, d \in R$ & $f(x) = \frac{(2a-1)x^3 + 5x^2 + 3x + (7c-d)}{(a+b)x^4 + 6x^2 + (b-3)x}$

If $\lim_{x \to \infty} f(x) = l_1$ and $\lim_{x \to 0} f(x) = l_2$ (l_1, l_2 are non zero and finite) then equation whose roots are l_1 and l_2 will be

(A) $7x^2 - 5x - 5 = 0$ (B) $42x^2 - x - 30 = 0$

(C) $7x^2 + 5x - 5 = 0$ (D) $42x^2 + x - 30 = 0$

53. If $f(x)$ is a decreasing function then the set of value of 'k', for which the major axis of the ellipse $\frac{x^2}{f(k^2 + 2k + 5)} + \frac{y^2}{f(k + 11)} = 1$ is the x-axis, is:

A) $k \in (-2, 3)$ B) $k \in (-3, 2)$

C) $k \in (-\infty, -3) \cup (2, \infty)$ D) $k \in (-\infty, -2) \cup (3, \infty)$

54. Let $x_1, x_2, \ldots x_n$ be n observations such that $\sum x_i^2 = 400$ and $\sum x_i = 80$.

Then a possible value of n among the following is:

A) 12 B) 9 C) 18 D) 15

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55. $g(x)=3+4x$, $gog(x)=g_2(x)$ $gogog(x)=g_3(x)$ then inverse of $g_n(x)$ is

(A) $4^{-n}\left(x+1-4^n\right)$ (B) $4^{-n}\left(x-1+4^n\right)$

(C) $4^{-n}\left(x+1+4^n\right)$ (D) $4^{-n}\left(4^n-x-1\right)$

56. If $|z|=3$, $\left|\frac{\arg(z)}{\arg(z+3)}\right|=$

(A) 3 (B) 2 (C) 4 (D) 1

57. Let S = {1, 2, 3, … , 9}. For k = 1, 2, … 5, let N_k be the number of subsets of S, each containing five elements out of which exactly k are odd. Then $N_1+N_2+N_3+N_4+N_5=$

(A) 125 (B) 252 (C) 210 (D) 126

58. $f(x)=\frac{x}{1+(\ln x)(\ln x)\ldots\infty}\ \forall x\in[1,3]$ is non- differentiable at x = k. Then the value of $\left[k^2\right]$ is_____. (where [.] represents the greatest integer function).

(A) 6 (B) 4 (C) 5 (D) 10

59. Value of $(1-\cot 22)(1+\tan 22)(1-\cot 23)x(1+\tan 22)$

(A) 8 (B) 4 (C) 16 (D) 64

60. A(–4, 0, 3) & B(14, 2, –5) then which one of the point lie on the bisector of angle bisector b/w $\overline{OA}$ & $\overline{OB}$

(A) (2, 2, 5) (B) (2, 11, 5) (C) (–3, –3, –6) (D) (1, 1, 3)

61. If 2 numbers are chosen from {0, 1, 2, …. 10} and probability. that, sum & their difference is multiple of 4 is $\frac{K}{55}$, then K is__

(A) 8 (B) 6 (C) 12 (D) 10

62. If $f:R\to R$ satisfies $f(x+y)=f(x)+f(y)$ for all $x,y\in R$ and $f(1)=7$, then $\sum_{r=1}^{n} f(r)$ is

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A) $\frac{7n}{2}$ B) $\frac{7(n+1)}{2}$ C) $7n(n+1)$ D) $\frac{7n(n+1)}{2}$

63. If the system of linear equations x + 2ay + az = 0, x + 3by + bz = 0, x + 4cy + cz = 0 has a non-trivial solution & a, b, c > 0 then the minimum value of $\left(\frac{a^3+c^3}{b^3}\right)$ is

(A) 1 (B) 0 (C) 2 (D) 3

64. $\vec{x}=3i-6j-k$, $y=i+4j-3k$, and $\vec{z}=3i-4j-12k$, then the magnitude of the scalar projection of $\vec{x}\times\vec{y}$ on $\vec{z}$ is:

(A) 12 (B) 15 (C) 14 (D) 13

65. If $p(x)=x^6-x^5-x^3-x^2-x$ & $Q(x)=x^4-x^3-x^2-1=0$ have roots $\alpha, \beta, \gamma, \delta$ then value of $P(\alpha)+P(\beta)+P(\gamma)+P(\delta)=$

(A) 2 (B) –1 (C) –3 (D) 0

66. $\int x^x\left((\ln x)^2+(\ln x)+\frac{1}{x}\right)dx=$

(A) $x^x\left((\ln x)^2-\frac{1}{x}\right)+c$ (B) $x^x(\ln x-x)+c$

(C) $x^x\frac{(\ln x)^2}{2}+c$ (D) $x^x\ln x+c$

67. The area of the region $\left\{(x,y):xy\le 8, 1\le y\le x^2\right\}$ is

(A) $8\log_e 2-\frac{14}{3}$ (B) $16\log_e 2-\frac{14}{3}$

(C) $16\log_e 2-6$ (D) $8\log_e 2-\frac{7}{3}$

68. Let X and Y be two arbitrary, 3 × 3, non-zero, skew-symmetric matrices and Z be an arbitrary 3 × 3, nonzero, symmetric matrix.

(1) $Y^3Z^4-Z^4Y^3$ (2) $X^{44}+Y^{44}$

(3) $X^4Z^3-Z^3X^4$ (4) $X^{23}+Y^{23}$

Then which of the following matrices is (are) skew symmetric?

(A) (2, 4) (B) (2, 3) (C) (1, 3) (D) (3, 4)

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69. If we consider only the principal values of the inverse trigonometric functions, then the value of $\tan\left(\cos^{-1}\frac{1}{5\sqrt{2}}-\sin^{-1}\frac{4}{\sqrt{(17)}}\right)$ is

(A) $\sqrt{29/3}$ (B) 29/3 (C) $\sqrt{3/29}$ (D) 3/29

70. In ΔABC, if $\cos A+\cos C=4\sin^2\frac{1}{2}B$, then a, b, c are

(A) H.P. (B) G.P. (C) A.P. (D) None of these

SECTION – II
(Numerical Value Answer Type)

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71. If $N=2^3\cdot 3^2$, S is sum of all divisors, R is sum of reciprocal of divisors of N and value of $\frac{S}{R}=2^a\cdot 6^b$ then a + b =

72. If $y=\left(\sin^{-1}x\right)^2+A\left(\cos^{-1}x\right)+B$ is represented by $\left(p-x^2\right)\frac{d^2y}{dx^2}-x\frac{dy}{dx}=q$, then value of p + q =

73. Two points A & B are taken on $xy=C^2$. let tangents at A, B cuts x-axis at C and D respectively, then ratio of the areas ΔOAC to that of ΔOBD is_____

74. $\lim\limits_{x\to 0}\frac{\ln\left(1+x+x^2\right)+\ln\left(1-x+x^2\right)}{\sec x-\cos x}$

75. If $f(x)=x^3+e^x$ and g(x) is inverse of f(x) value of $g^1(1)=$

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